



- **First Law.** An object at rest stays at rest unless acted on by an external force. An object in motion continues to travel with constant speed in a straight line unless acted on by an external force. It is also known as the law of inertia. Inertia can be viewed as the resistance to any change in an object's velocity. Or the tendency of a body to keep moving once it is set in motion.
- **Second Law.** The acceleration of an object is directly proportional to the net force acting on it, and the reciprocal of the mass of the object is the constant of proportionality. Thus, $\vec{F}_{\text{net}} = m\vec{a}$, where $\vec{F}_{\text{net}} = \sum \vec{F}$.
- **Third Law.** When two bodies interact, the force $\vec{F}_{BA}$ exerted by object B on object A is equal in magnitude and opposite in direction to the force $\vec{F}_{AB}$ exerted by object A on object B. Thus, $\vec{F}_{BA} = -\vec{F}_{AB}$.

These laws can not be proved, therefore they are considered as fundamental principles. These three laws are valid only in the inertial frames of reference!

Superposition of forces principle: Tested by experiments, any number of forces applied at a point on a body have the same effect as a single force equal to the vector sum of the forces. In addition, a single force can be decomposed into different components.

Range of validity: Newtonian mechanics does not apply to all situations. For fast-moving objects (close to c), we must replace Newtonian mechanics with Einstein's special theory of relativity. For small particles, we must use quantum mechanics instead. Newtonian mechanics is the special case of these two more comprehensive theories.

2.1 Newton's First Law of Motion

- **First Law.** An object acted on by no net force moves with constant velocity and zero acceleration.
- It is also known as the law of inertia.
- **Inertial frames of reference:** the frame in which Newton's first law is valid. If no forces act on an object, any reference frame for which the acceleration of the object remains zero is an inertial reference frame. Inertial frame should be a frame with no acceleration with respect to other inertial frames. There are many inertial frames, since any frame B moving with a constant velocity with respect to an inertial frame A is also an inertial frame of reference. The velocity in frame B is $\vec{v}_{p/B} = \vec{v}_{p/A} + \vec{v}_{A/B} = \vec{v}_{p/A} - \vec{v}_{B/A}$ which is a constant as long as $\vec{v}_{p/A}$ and $\vec{v}_{B/A}$ are constants. The earth is approximately an inertial frame, due to the small acceleration associated with its rotations.

Inertia is the resistance to any change in an object's velocity. Or the tendency of a body to keep moving once it

is set in motion as shown above.

2.2 Newton's Second Law of Motion

$$\vec{F} = m\vec{a}, \quad \text{vector equation} \quad (1)$$

$$F_x = ma_x, \quad F_y = ma_y, \quad F_z = ma_z, \quad \text{components.} \quad (2)$$

What is mass?

$m = |F|/a$ tells us that the ratio $|F|/a$ is a constant for a given body. We call this ratio the **inertial mass**, or simply the mass. Mass is a quantitative measure of inertia. The greater the mass, the more a body "resists" being accelerated (change of motion).

Relativity

In Einstein's special relativity, there is mass-energy equivalence (relation) $m = E/c^2$. ("Does the inertia of a body depend upon its energy-content?") In general relativity, the equivalence principle is the equivalence of gravitational and inertial mass.

- **Heads up:** first application of vectors in physics.
- $F = 0$ gives constant velocity.
- For the same force, the ratio of the masses is the the inverse of the ratio of the accelerations $\frac{m_1}{m_2} = \frac{a_2}{a_1}$. It is harder to accelerate a heavy object than a light object.
- Mass is not the same as weight, although we treat them the same in our daily life. Weight is a force $W = mg$ with $g = 9.8\text{N/kg} = 9.8\text{m/s}^2$.
- Non-inertial frame: Imagine we drop two objects A and B simultaneously (at the same time) and let them free fall. Choosing the object A as the reference frame, we find that object B is at rest relative to object A. However, $F_B = mg$ is not zero, which means that Newton's first and second laws are no longer valid. This is due to the reason that object A is accelerating and it is not an inertial frame.

2.3 Newton's Third Law of Motion

If body A exerts a force on body B (also called an action), the body B exerts a force on body A (a reaction). These two forces have the same magnitude but are opposite in direction. These two forces act on different bodies. The action and reaction are two opposite forces, which often are referred as an action-reaction pair.

3 Applying Newton's Laws

Free body diagram

A free body diagram is a diagram showing the chosen